

# EMEL ÖRKEN

Full Stack AI Engineer | FastAPI · Django · React | RAG · Agent Systems & AI Products

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## PROFESSIONAL SUMMARY

Full Stack AI Engineer who builds production-grade backend systems, integrates AI into those systems, and deploys with Docker. Designed and stood up a 3-container microservice architecture from scratch; built RAG chatbots and multi-agent systems with LangChain and LangGraph; wrote production REST APIs with FastAPI and Flask. Capable of owning both the infrastructure layer and the intelligence layer independently.

## EXPERIENCE

### AI Engineer · MANK Technologies

Jul 2025 – Aug 2025

- Cleaned and processed 70+ hours of raw audio into WAV+SRT format for model training; built a text normalization pipeline automating number-to-word conversion.
- Developed a fully functional conversational AI product for credit card sales workflows using Ultravox and prompt engineering, delivered from scratch in two weeks.

**Stack:** Python · Whisper · Ultravox · Prompt Engineering · Ffmpeg · WER/CER Evaluation

### Software Engineer Intern · TURK AI

Feb 2025 – Jun 2025 · Ankara

- Designed a 3-container Docker microservice system (RabbitMQ + PostgreSQL + MinIO) for real-time INTERPOL Red Notice data tracking; built a live Flask dashboard with filtering, sorting, and pagination.
- Transcribed YouTube videos with Whisper, embedded content into a vector database, and built a RAG chatbot with multi-collection semantic search.

**Stack:** Python · Flask · PostgreSQL · RabbitMQ · Docker · Docker Compose · WebSocket · LangChain · Whisper · MinIO · Git

### AI Engineer Intern · CBOT

Sep 2024

- Built a document-based RAG pipeline with FAISS and LangChain; implemented PDF/text extraction, HTML parsing, and vector DB ingestion with clean Python abstractions.
- Integrated HuggingFace Transformers for LLM usage; systematically improved query quality and model outputs through prompt engineering.
- Delivered rapid AI prototypes with Gradio and Streamlit; prototyped multimodal AI applications with CLIP and BLIP.

**Stack:** Python · LangChain · FAISS · HuggingFace · Transformers · Whisper · PostgreSQL · Gradio · Streamlit · Git

## NOTABLE PROJECTS

### INTERPOL Real-Time Data Tracking System — *Docker · RabbitMQ · WebSocket · Flask*

3-container Docker microservice · RabbitMQ message queue · PostgreSQL + Redis + MinIO · WebSocket real-time streaming · Flask REST API · BeautifulSoup scraping · rate-limit & retry pattern · live filterable dashboard

### YouTube RAG Chatbot — *LangChain · Vector DB · PostgreSQL · MinIO*

Whisper transcription → chunking → vector embedding → multi-collection semantic search REST API · MinIO video storage · session history UI · end-to-end RAG pipeline

### Telegram Product Research Agent — *LangGraph · Python*

LangGraph multi-agent system · parallel search nodes · BeautifulSoup scraping · human-in-the-loop Telegram inline button approval mechanism

### Multi-Agent Article Writing System — *LangGraph · Python*

Supervisor architecture · planner, researcher, writer and editor agents · fully automated flow from web research to edited article

### Voice AI Sales Demo — *Ultravox · Prompt Engineering*

Conversational AI assistant for credit card customer service · Ultravox + prompt engineering · delivered from scratch in two weeks · prompt optimization and conversation flow design

TECHNICAL SKILLS

**Backend:** FastAPI · Flask · Django · REST API · WebSocket · Async Programming  
**Frontend:** React · HTML · DataTable.js  
**Languages:** Python · SQL (DDL, DML, Stored Procedure) · C# · Go (basic)  
**Database:** PostgreSQL · Redis · SQLite · Microsoft SQL Server · MinIO  
**Message Queue:** RabbitMQ · async task management · rate-limit & retry pattern  
**DevOps:** Docker · Docker Compose · Kubernetes · Git · Linux · Railway · Alembic · JWT  
**LLM & AI:** LangChain · LangGraph · FAISS · Vector DB · Ollama · OpenAI API · Prompt Engineering  
**AI Products:** RAG Systems · Multi-Agent Architecture · Conversational AI · Whisper Pipeline  
**Web Scraping:** BeautifulSoup · Playwright · HTML Parsing  
**Deep Learning:** PyTorch · TensorFlow · Hugging Face · YOLOv11/v12 · OpenCV · CVAT

EDUCATION

|                                                                         |                     |
|-------------------------------------------------------------------------|---------------------|
| <b>B.Sc. Software Engineering</b> · Atatürk University, Erzurum, Turkey | Sep 2021 – Jun 2025 |
| GPA: 3.26 / 4.00                                                        |                     |
| <b>Selçuklu Science High School</b>                                     | Oct 2016 – Jun 2020 |
| GPA: 95.2 / 100 · Ranked 2nd in school                                  |                     |

AWARDS & ADDITIONAL

TÜBİTAK High School Research Projects — Regional Acceptance (2019) · TEV Full Scholarship (2022–2025) · TEV Mentorship Program Volunteer

REFERENCES

**Barış Ünver** · Founder & CEO, TURK AI · +90 505 820 23 03 (Reference Letter Available)  
**Tüncer Keleş** · HR Director · +90 532 603 49 93  
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